

When needed, Show your work

Quiz 1 September 4<sup>th</sup> 2025 Chem 1127

Name (First, Last):

Key

Note the quiz is front and back

Points: \_\_\_\_/16

1. Graphite is pure carbon (C). Identify graphite as a(n) C Circle the correct term (1 pt)  
a. homogeneous mixture    b. heterogeneous mixture    c. element    d. compound

2. Briefly define the term heterogeneous mixture and provide an example of it. Your definition should incorporate the bolded prefix of the first word. (1.5 pt) - Damascus Steel  
- Coffee + Milk  
- Oil + Vinegar  
→ Mixture of 2 or more compounds/elements that has different parts throughout

3. Circle the property below that is not a physical property. (1 pt)  
a. Melting Point    b. Density    c. Toxicity    d. Color

4. Density is an intensive or extensive property. Identify it and explain your identification. (1 pt)

→ Does not depend on the sample

5. Clearly identify how many significant figures are in the two values below. (1 pt)

a.  $2.60 \times 10^{-4}$

b. 0.5025

3

4

6. My car has a 13 gal tank, but I am filling up in Canada, where the average price of gas is 1.86 CAD/1 L gas. How much, in USD, does it cost me to fill up my tank in Canada? (1.5 pt)

1 gal = 3.79 L

1 USD = 1.40 CAD

$$\frac{13 \text{ gal}}{1} \cdot \frac{3.79 \text{ L}}{1 \text{ gal}} \cdot \frac{1.86 \text{ CAD}}{1 \text{ L}} \cdot \frac{1 \text{ USD}}{1.40 \text{ CAD}} = 65.46 \text{ USD}$$

- a. Write a conversion factor (fraction or ratio) for the above question. (1 pt)

$$\frac{1 \text{ USD}}{1.40 \text{ CAD}}$$

- b. Identify one of the given numbers above as a measured number/value. (0.5 pt)

13 gal

7. Under standard conditions, what is the density (g/mL) of water to one significant figure. (0.5 pt)

a. 100 g/mL

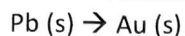
b. 1 g/mL

c. 0.0001 g/mL

d. 24 g/mL

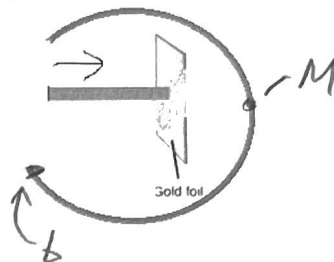
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8. Is this a chemical reaction per Dalton's Atomic Theory? Why or why not? (1 pt)



No. This would require a Pb to be lost. Atoms are conserved.

9. In the image of the Gold Foil Experiment, draw a point where most of the incoming alpha particles (flat rectangle) hit the circular detector after shining on the gold foil and clearly label it "M". (0.5 pt)



- a. This result indicated that atoms were mostly **filled** or **empty** space, Circle the correct bolded word. (0.5 pt)

- b. Identify a second point, labeled "b", on the detector which indicated the nucleus of an atom. You should explain your answer if it seems ambiguous. (1 pt) Hit massive, positive Nucleus, deflected Backwards,

10. Opposite charges Attracted. Fill in the blank with a correct verb/word. (0.5 pts)

11. ~ End of Class Tuesday. An ion,  $^{32}\text{S}^{2-}$ , contains how many protons (p), neutrons (n), and electrons (e) and identify this as an **anion** or **cation**. (2 pt)

protons (p)

neutrons (n)

electrons (e)

**anion** or **cation**.

15  
 16 → Mass # A is 32 = p + n  
 15 + 16  
 → Atomic #, Z from P.T

12. For the following blanks fill them in with proton (p), electron (e), or neutron (n). Note, one or more of the blanks should be left blank. (1.5 pt)

- a. p & n are the heaviest or of equally heavy mass.  
 b. e & e is (are) negatively charged.  
 c. p & n reside **inside** the nucleus of an atom.

Key

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Show your work

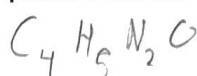
Points: \_\_\_\_/16<sub>17.5</sub>

1. Given the structure of caffeine shown here, clearly write its molecular and empirical formula. They may be the same thing. (1.5 pt)

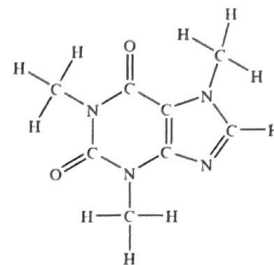
Molecular Formula



Empirical Formula



remove  
Greatest common factor of 2.



2. ~Aktiv HW. An unknown element X has the following isotopes:  $^{10}X$  (20.00% abundant, atomic mass = 10.0 amu) and  $^{13}X$  (80.00% abundant, atomic mass = 13.0 amu). What is the average atomic mass in amu of X? Express your answer to three significant figures. (1 pt)

$$A \text{ AM} = 0.2 \times 10.0 + 0.8 \times 13.0 = 12.4 \text{ amu}$$

3. Circle the below element(s) that is (are) **diatomic** in their elemental form. (1 pt)

H

Li

O

Fe

- a. For one of the diatomic elements, write the diatomic formula below. (0.5 pt)



2 atoms



4. How many grams of  $C_2H_4$  are in 1.6 mol of  $C_2H_4$ ? (1 pt) MM

$$\frac{1.6 \text{ mol } C_2H_4}{1} \cdot \frac{28 \text{ g } C_2H_4}{1 \text{ mol } C_2H_4} = 44.8 \text{ g } C_2H_4$$

- a. The problem above requires you to identify the molar mass (MM) of  $C_2H_4$ . Do so and write it as a conversion factor with both units. If it is in the main part, that is fine. (1 pt)

$$\text{MM} \quad \begin{matrix} 2 \times 12 \\ C_2 \end{matrix} + \begin{matrix} 4 \times 1 \\ H_4 \end{matrix} = \frac{28 \text{ g } C_2H_4}{1 \text{ mol } C_2H_4}$$

- b. How many molecules of  $C_2H_4$  are present in 1.6 mol of  $C_2H_4$ ? (0.5 pt)

$$\frac{1.6 \text{ mol } C_2H_4}{1} \cdot \frac{6.02 \times 10^{23} \text{ molecules } C_2H_4}{1 \text{ mol } C_2H_4} = 9.6 \times 10^{23} \text{ molecules } C_2H_4$$

5. ~A2.73. How many hydrogen atoms are present in one molecule ("formula unit") of ammonium benzoate,  $(C_6H_5COONH_4)$ ? (0.5 pt)

$$5 + 4 \rightarrow 9 \quad \frac{9 \text{ H atoms}}{1 \text{ molecule } C_6H_5COONH_4}$$

- a. ~A2.73. How many hydrogens would be present in three molecules of it? (0.5 pt)

$$\frac{3 \text{ molecules } C_6H_5COONH_4}{1} \cdot \frac{9 \text{ atoms H}}{1 \text{ molecule } C_6H_5COONH_4} = 27 \text{ atoms H}$$

6. If the wavelength of electromagnetic radiation increases, how does the frequency & energy change?

Do they stay constant, increase, or decrease? (1 pt)

frequency

energy

$$\lambda \uparrow \quad c = \lambda \cdot \nu \quad \begin{matrix} \uparrow & \downarrow \end{matrix} \quad \rightarrow \quad \begin{matrix} \text{Decrease} \\ E = h \cdot \nu \\ \downarrow & \downarrow \end{matrix}$$

Decrease

7. A radio wave has a frequency of 275 GHz, <sup>V</sup> <sup>W</sup> what is the wavelength in meters of these waves? (1.5 pt)

$$\frac{275 \text{ GHz}}{1} \cdot \frac{10^9 \text{ Hz}}{\text{GHz}} = 275 \times 10^9 \text{ Hz} = \nu = 275 \times 10^9 \frac{1}{\text{s}}$$

$$\frac{c}{\nu} = \lambda = \frac{3 \times 10^8 \text{ m/s}}{275 \times 10^9 \frac{1}{\text{s}}} = 0.00109 \text{ m} \quad \text{or} \quad 1.09 \times 10^{-3} \text{ m}$$

- a. How can the units of Hz be rewritten in terms of seconds (s)? Getting the main part earns you these points. (0.5 pt)

$$\text{Hz} = \frac{1}{\text{s}} = \text{s}^{-1}$$

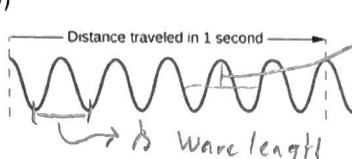
- b. What is the energy of 1.3 moles of the above photons given the frequency. (1 pt)

1st  $E = h \cdot \nu$   $2nd \rightarrow 1.3 \text{ mol} \cdot \frac{6.02 \times 10^{23}}{1 \text{ mol}} \cdot \frac{1.82 \times 10^{-22} \text{ J}}{\text{photon}} = 1.43 \times 10^2 \text{ J}$

$$E = 6.63 \times 10^{-34} \text{ J} \cdot \text{s} \cdot 275 \times 10^9 \frac{1}{\text{s}} = 1.82 \times 10^{-22} \text{ J / photon}$$

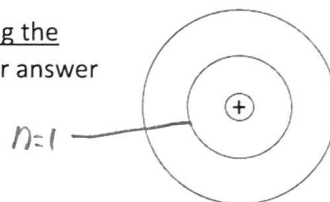
$\rightarrow$  Energy of 1 photon

8. Identify the frequency and wavelength and amplitude, given this image that shows how many waves there are in 1 second. (1.5)



Frequency = 6 Hz =  $\frac{6 \text{ waves}}{1 \text{ second}}$

9. Given this image of the Bohr model, with the innermost circle indicating the nucleus, identify the orbit corresponding to  $n = 1$ , you may explain your answer if you feel it necessary. (0.5 pt)



- a. Is the above orbit the lowest energy orbit? Yes or No (0.5 pt)

- b. Draw in the orbit for  $n = 1.5$ , or circle "not allowed". (0.5 pt)

$\rightarrow n$  only equals 1, 2, 3, 4, ...

10. ~ A3. For a hydrogen atom, calculate the frequency of light (in  $1/\text{s}$  or  $\text{s}^{-1}$ ) that would be emitted for the orbital transition of  $n(\text{initial}) = 2$  to  $n(\text{final}) = 1$ . Submit an answer to 3 significant figures. The Rydberg constant is  $1.09678 \times 10^7 \text{ m}^{-1}$ . (1.5 pts)

$$\nu = \frac{R_H}{h} \left[ \frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \rightarrow \nu = \frac{2.18 \times 10^{-18} \text{ J}}{6.63 \times 10^{-34} \text{ J} \cdot \text{s}} \cdot \left[ \frac{1}{1^2} - \frac{1}{2^2} \right] = 2.47 \times 10^{15} \frac{1}{\text{s}}$$

$E = h \cdot \nu$  if  $\nu \uparrow \rightarrow$  Energy/speed of electrons  $\uparrow$

11. In the photoelectric effect, if the frequency of light shone on a metal increases, what will happen?

Circle it (1 pt)

- a. Ejected electrons ( $e^-$ ) will be faster  
b. There will be more ( $e^-$ )  
c. Ejected ( $e^-$ ) will be slower  
d. There will be fewer ( $e^-$ )  
e. No change

12. In our discussion on the Law of Conservation of Energy, this man was mentioned. Spell his last name correctly. Alternatively, what alternative history comic book was he a character in? Last Alternatively, explain the term **orbital** or what  $|\psi|^2$  means. (0.5 pts)

Feynman / Manhattan projects

$\rightarrow$  tells us probability that electron is there.  
 $\rightarrow$  space where  $e^-$  likely is

